

LOM, Bulgaria

WMO: 155110

Lat: 43.817N

Lon: 23.250E

Elev: 33

StdP: 100.93

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.1	-12.1	1.3	-10.0	-9.6	1.7	-7.3	12.4	6.7	10.5	4.6	1.6	50	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.6	24.3	32.0	23.7	30.5	23.1	25.5	31.4	24.6	30.4	23.7	29.4	2.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	29.6	22.7	17.5	28.5	21.8	16.6	27.5	78.5	31.5	74.6	30.5	71.1	29.4	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	9.7	8.0	DB	-14.1	35.8	3.5	1.9	-16.6	37.2	-18.7	38.3	-20.7	39.3	-23.2	40.7	(4)
(5)			WB	-13.9	27.3	3.5	1.5	-16.4	28.4	-18.4	29.3	-20.4	30.2	-22.9	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	0.5	2.7	7.7	13.1	18.4	22.3	24.5	24.0	19.0	12.8	6.6	1.2	(6)
(7)		DBStd	9.34	4.66	5.20	4.44	3.28	3.16	2.93	2.55	2.70	3.19	3.96	3.86	4.34	(7)
(8)		HDD10.0	1017	294	208	97	11	0	0	0	0	18	115	273		(8)
(9)		HDD18.3	2612	553	438	329	159	39	4	1	1	30	176	352	531	(9)
(10)		CDD10.0	2037	1	3	26	103	262	368	449	434	271	106	13	1	(10)
(11)		CDD18.3	589	0	0	0	2	42	122	192	177	51	4	0	0	(11)
(12)		CDH23.3	4979	0	0	1	17	294	1033	1784	1504	316	29	1	0	(12)
(13)		CDH26.7	1796	0	0	0	2	49	339	731	595	76	3	0	0	(13)
(14)	Wind	WSAvg	2.4	2.3	2.7	2.9	2.6	2.6	2.5	2.2	2.2	2.1	2.1	2.2	(14)	
(15)	Precipitation	PrecAvg	561	38	37	37	49	62	54	56	43	46	49	37	52	(15)
(16)		PrecMax	868	83	74	98	99	115	105	131	208	120	143	95	117	(16)
(17)		PrecMin	309	6	3	7	1	20	17	5	2	5	5	1	11	(17)
(18)		PrecStd	129	21	22	23	23	26	26	37	44	33	36	26	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.9	21.7	25.6	30.1	34.2	36.7	35.7	31.2	26.7	19.8	14.3	(19)
(20)			MCWB	9.6	11.1	13.4	17.4	21.0	24.4	23.9	24.1	22.0	19.3	14.2	10.1	(20)
(21)		2%	DB	10.9	14.4	19.2	22.6	27.9	31.8	33.8	33.4	28.9	23.2	16.2	10.0	(21)
(22)			MCWB	7.9	9.8	12.5	15.5	20.2	23.6	24.1	24.5	20.9	18.1	12.7	7.6	(22)
(23)		5%	DB	8.5	11.9	16.9	20.6	26.5	30.2	32.0	31.8	26.8	20.7	13.8	7.8	(23)
(24)			MCWB	6.0	8.3	11.4	14.5	19.6	22.8	23.7	23.8	20.1	16.7	11.6	6.2	(24)
(25)		10%	DB	6.4	9.6	14.7	19.0	24.9	28.6	30.5	30.1	24.9	18.6	12.0	6.4	(25)
(26)			MCWB	4.6	6.8	10.1	13.7	18.9	22.1	23.3	23.0	19.0	15.3	10.3	5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	11.8	14.2	18.5	22.6	26.0	26.5	27.2	23.1	20.2	15.2	10.2	(27)
(28)			MCD B	13.7	16.2	19.8	23.5	27.6	31.9	32.6	32.3	28.7	24.8	18.4	13.6	(28)
(29)		2%	WB	7.9	10.0	12.8	16.5	21.2	24.6	25.5	25.6	21.8	18.5	13.4	8.1	(29)
(30)			MCD B	10.7	13.9	18.3	21.3	26.5	30.4	31.4	31.4	27.8	22.2	15.7	9.7	(30)
(31)		5%	WB	6.1	8.5	11.5	15.2	20.2	23.5	24.6	24.5	20.7	17.1	11.9	6.5	(31)
(32)			MCD B	8.1	11.7	16.3	19.4	25.3	29.1	30.5	30.3	25.6	20.4	13.4	7.5	(32)
(33)		10%	WB	4.8	7.0	10.3	14.2	19.2	22.5	23.8	23.5	19.6	15.6	10.5	5.2	(33)
(34)			MCD B	6.3	9.4	14.2	18.0	24.1	27.8	29.5	29.0	24.0	18.2	12.0	6.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	6.5	8.4	9.0	9.7	10.1	10.7	9.6	7.9	5.4	4.6	(35)	
(36)			MCD BR	8.9	11.0	12.8	12.1	12.5	12.6	13.2	13.2	12.7	11.4	9.1	7.1	(36)
(37)			MCWBR	6.3	7.3	7.5	6.5	6.2	6.3	6.7	6.7	6.5	7.0	6.0	5.1	(37)
(38)		5% WB	MCD BR	8.2	10.4	12.0	10.8	11.4	11.9	12.1	11.7	10.8	7.9	6.3	(38)	
(39)			MCWBR	6.0	7.1	7.1	6.2	6.2	6.5	6.6	6.7	6.5	7.0	5.5	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.377	0.417	0.479	0.458	0.454	0.465	0.482	0.450	0.406	0.374	0.348	(40)	
(41)		taud	2.286	2.202	2.155	2.003	2.132	2.185	2.156	2.104	2.157	2.270	2.311	2.308	(41)	
(42)		Ebn at Noon	754	794	811	788	819	823	808	777	769	754	717	713	(42)	
(43)		Edh at Noon	86	111	133	167	151	144	147	149	131	103	84	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.43	2.25	3.39	4.61	5.63	6.32	6.40	5.70	4.10	2.67	1.54	1.19	(44)	
(45)		RadStd	0.22	0.32	0.41	0.52	0.49	0.49	0.42	0.47	0.34	0.37	0.21	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.56	N/A	N/A	N/A	+0.40	+0.40	N/A	-152	+77	N/A

Nomenclature: See separate page